

An MDP and Genetic-Algorithm Study of the Blood Donation Tailoring Problem

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Why blood donation tailoring?

- A blood organization has a fixed pool of repeat donors.
- Each donation type collects a different mix of **RBC, platelets, plasma**.
- Products are perishable; demand varies by period.
- Each donation locks the donor out for a type-dependent **deferral** window.
- **Decision:** for each donor and each day pick a donation type (or NONE) that minimizes donation + holding + disposal cost.
- Following Ozener, Ekici, Coban (2019).

Sets, parameters, decision variables

Sets: donors K , types $I = \{WB, SP, DP, RP, RPP, NONE\}$, products $J = \{RBC, \text{platelets}, \text{plasma}\}$, periods $T = \{0, \dots, L-1\}$.

Parameters: a_{ij} collection, c_i donation cost, h_j holding, d'_j disposal, sh_j shelf life, E_j donor cap, s_{ij} deferral, d_{jt} demand.

Decision variables: $x_{kit} \in \{0, 1\}$, $I_{jt} \geq 0$, $P_{jt} \geq 0$.

MIP formulation

$$\min \sum_{k, i, t} c_i x_{kit} + \sum_{j, t} h_j I_{jt} + \sum_{j, t} d'_j P_{jt}$$

subject to: assignment, deferral, inventory balance, donor cap, shelf-life FIFO.

Implementation: Gurobi 11; exact when tractable; size-limited license caps the largest instances.

MDP reformulation

$$s_t = (t, (cd_k)_k, (u_{kj})_{kj}, (inv_j)_j)$$

Action $a_t = (i_k)$ feasible iff $cd_k > 0 \square i_k = \text{NONE}$ (deferral) and $u_{kj} + a_{i_k j} \leq E_j$ (donor cap).

Transition: deterministic -- collect, age, serve FIFO, expire, update cooldown and used.

Type: deterministic, fully observable, undiscounted, finite horizon.

Bellman equation

$$V_t(s) = \min_{a \in \mathcal{A}(s)} \{c_t(s, a) + V_{t+1}(f(s, a))\}, \quad V_L(s) = 0$$

$V_t(s)$: minimum remaining cost from state s at period t .

$c_t(s, a)$: stage cost (donation + end-of-period holding + disposal of aged-out units).

Sanity check ($N=2$, $L=4$): MIP = MDP-DP = **1457.00**; the MDP explores 44 states.

Genetic algorithm

- **Encoding:** chromosome of length $N L$; gene $k L+t$ = donation type of donor k at period t .
- **Repair-on-evaluate:** walk through periods and substitute NONE for any gene that violates cooldown or donor cap; infeasibility penalized $10^6(L-t^*)$.
- **Operators:** tournament ($k=3$) + elitism, two-point crossover (rate 0.8), per-gene resampling mutation (rate $1/NL$).
- **Warm-start:** seed initial population with myopic-greedy and per-type uniform schedules; rest random.
- **Defaults:** pop 80, generations 150 (not tuned per instance).

Experiment setup

Grid: $N \in \{3, 5, 8\}$, $L \in \{8, 14, 21\}$, demand $\in \{\text{constant, increasing, random}\}$, instance seed = 7. 27 instances total.

GA seeds: $\{1, 2, 3\}$ per instance $\Rightarrow$ **81 GA runs + 81 baseline runs.**

Methods: MIP (Gurobi 11, 120 s limit), Rule-of-Thumb (uniform single type), Myopic Greedy, GA. MDP-DP only on the $N=2$, $L=4$ sanity instance.

Hardware: single Windows 11 laptop; runtimes are wall-clock.

Sanity check on $N=2$, $L=4$

Method	Objective	Runtime [s]
MIP	1457.00	0.004
MDP-DP	1457.00	0.001
Rule-of-thumb	1460.40	0.000
Myopic greedy	infeasible	0.000
GA	1457.00	0.358

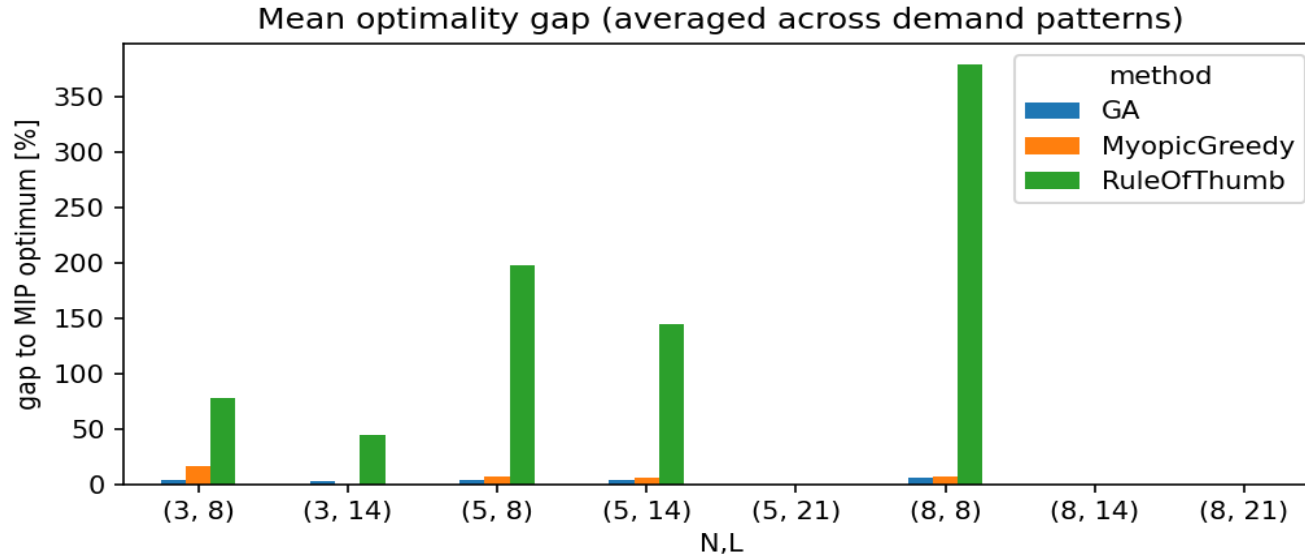
All exact methods agree on 1457.00. Greedy is infeasible (puts both donors on cooldown simultaneously). GA recovers the optimum.

Main results -- selected instances, constant demand

N	L	MIP	GA (mean)	Greedy	RoT
3	8	2,151.6	2,299.9	2,643.6	4,412.4
3	14	3,661.2	3,955.0	--	6,666.4
5	8	2,151.6	2,189.7	2,371.6	7,406.4
5	14	3,661.2	3,763.4	3,991.2	11,258.0
5	21	--	5,612.2	5,908.5	--
8	8	2,151.6	2,335.9	2,371.6	11,914.4
8	14	--	3,871.8	3,991.2	18,132.4
8	21	--	5,673.3	5,908.5	--

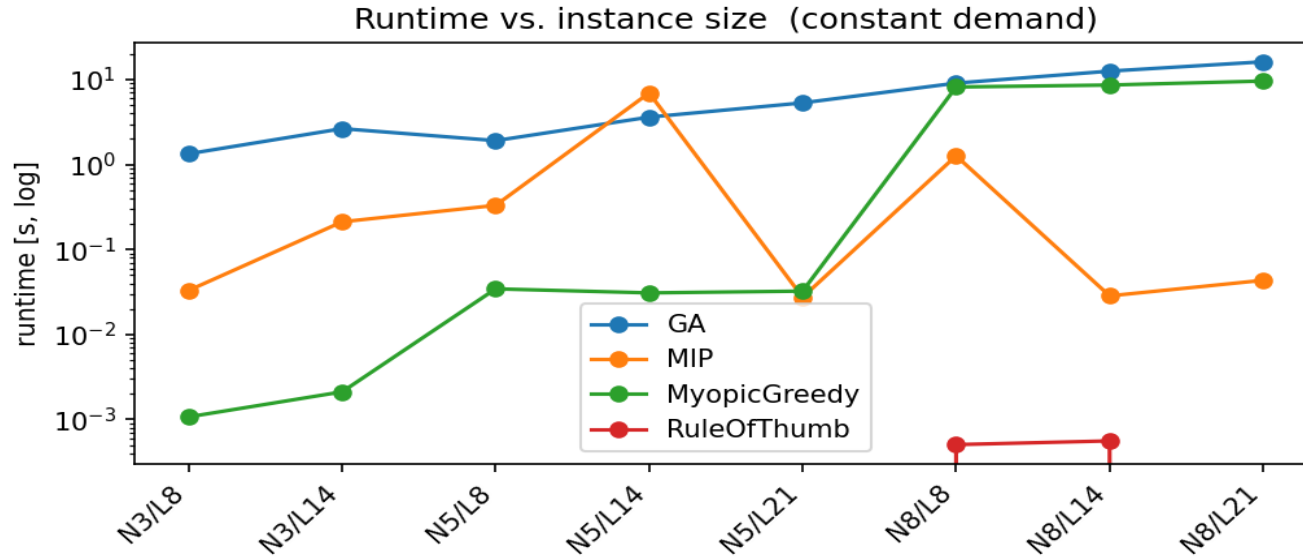
GA stays within **3-6%** of MIP. RoT deteriorates rapidly with N . "--" = MIP license cap or method infeasible.

Optimality gap



Gap above the MIP optimum, mean over demand patterns. GA: **3-6%**. Greedy: 6-16% when feasible. RoT: $\geq 45\%$, up to **379%** at $N=8$.

Runtime scaling



Wall-clock seconds, log scale. GA scales smoothly with NL ; the MIP times out / hits the license cap on the largest instances.

What we learned

- **Rule-of-thumb** is robust but uniformly dominated -- a single donation type cannot match the demand mix.
- **Myopic greedy** is fragile under tight cooldowns: it can stall the donor pool simultaneously (5 of 24 instances infeasible).
- **GA** tracks the MIP optimum within a single-digit gap on small instances, and remains the only feasible exact-quality method when the MIP exceeds the license size limit.
- **The MDP reformulation is the bridge:** its repair logic and cost computation are reused inside the GA fitness.

Conclusion and future work

Findings

- BDTP is naturally a sequential decision problem; the MDP view absorbs deferral and donor-cap constraints into the action set.
- Population search with constraint repair is a workable scalable alternative to the MIP.

Limitations

- Single instance seed per (N , L , demand) combination.
- Free Gurobi license caps the largest instance for the MIP.
- Myopic greedy is intentionally simple.

Future work: stochastic demand, approximate dynamic programming, GA operator tuning across a larger grid, integration with real donor cohort data.

References

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- Eiben, Smith. *Introduction to Evolutionary Computing*. Springer, 2015.
- Fortin et al. *DEAP: Evolutionary algorithms made easy*. JMLR, 2012.

Thanks. Questions?